MIL-V-23300C

25 March 1976
SUPERSEDING
MIL-V-23300B
2 May 1969

MILITARY SPECIFICATION

VALVES, REGULATING, FLUID PRESSURE, CRANKCASE, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compressor crankcase pressure regulating valves for refrigeration systems using refrigerant-12 and having input powers of 10 horsepower or less.

1.2 Classification. The valves shall be of the following sizes and styles, as specified (see 6.2).

SIZES:

	Minimum capacity Btu/hr. evaporator temperature		Female solder type connections
Sizes	Minus 10°F. at 1 psi max. pressure-drop	Plus 20°F. at 2 psi max. pressure-drop	Inlet and outlet size (inches)
Size 7	6,000	12,000	7/8
Size 8	9,000	18,000	7/8
Size 9	12,000	24,000	1-1/8

STYLES:

Style 1 - Angle
Style 2 - Straight-Through

FSC 4820/4130

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein) Packaging Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MS 17846 - Valves, Regulating, Fluid Pressure, Crankcase, Refrigerant-12.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard product. The valves delivered under this specification shall be the current standard product of the manufacturer, except for changes necessary to meet specification requirements.

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for valves provided the completed item complies with all provisions of this specification.

3.3 Design and construction. The design and construction of the valves shall comply with the capacity rating, pressure drop, and temperature conditions indicated in 1.2 and MS 17846, when tested as specified in 4.3.1. The valves shall have a minimum operating range of 0 to 35 pounds per square inch gage. The valves shall withstand a minimum pressure of 140 psig without external leakage, and, for bursting strength, the valve shall withstand a minimum pressure of 350 psig without failure when tested as specified in 4.3.2.

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3.4 Finish. The finish shall be the manufacturer's standard finish.

3.5 Marking. The manufacturer's standard identification and inlet and outlet flow direction shall be placed on all valves according to the manufacturer's standard practice.

3.6 Workmanship. The crankcase pressure regulating valves shall be free of burrs, rough edges or slivers, and malformation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of components and materials. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all valves of one size and style offered for inspection at one time. The sample unit shall be one completely fabricated valve.

4.2.2.1 Visual examination. The completely assembled valve shall be examined for the defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect
Finish	Not manufacturer's standard finish

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TABLE I. Classification of defects (cont'd)

Examine	Defect
Design, construction and workmanship general (applicable to all components)	Burrs, rough edges or slivers Any part missing, fractured, malformed, dented or creased
Assembly	Component not connected or joined properly
Marking	Missing, incomplete, not legible

4.2.2.2 Dimensional examination. Examination shall be made of the valve to determine compliance with dimensional requirements. Any deviation from the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. Tests specified in 4.3.1 and 4.3.2 shall be performed using an inspection level of S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.2.3 Inspection of preparation for delivery. Preservation and packaging, packing, and marking shall be inspected in accordance with the quality assurance provisions in MIL-V-3.

4.3 Tests.

4.3.1 Performance and operational test. The valves shall be tested pneumatically to determine compliance with design requirements of 3.3. The pneumatic pressure shall be applied to simulate the conditions and rating of the valve. The valve internal operation shall be tested in conjunction with pressure drop to determine compliance with the minimum operating range pressures in 3.3. Nonconformance to paragraph 3.3 constitutes failure of this test.

* 4.3.2 Leak and failure test. The valves shall be subjected to the minimum leak test pressure of 140 psig and minimum strength test pressure of 350 psig to determine compliance with requirements in 3.3. The leak and failure test shall be performed using refrigerant backed by pneumatic, or hydrostatic pressures if manufacturer's test method requires. The leak test

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pressures shall be maintained a minimum of 10 minutes without external leakage. The failure test pressure shall be maintained a minimum of 1 minute without failure. Nonconformance to paragraph 3.3 constitutes failure of this test.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-V-3.

6. NOTES

6.1 Intended use. The crankcase pressure regulating valves covered by this specification are intended for use in refrigerating systems where compressor electric motor or engine drive protection is required because of:

(a) High starting loads.
(b) Surges in suction pressure.
(c) High suction pressure caused by hot gas defrost.
(d) Prolonged operation at excessive suction pressures.
(e) Low voltage and high suction pressure conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and styles of valves required (see 1.2).

6.3 The outside margins of this document have been marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationahip to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4130-0165

Review activities:

Army - CE
DSA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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